BILL-TO-TALLY AUTOMATION SOLUTION

Purchase Invoice Processing for TallyPrime

Prepared for	Ashan Tech Labs
Workflow	Upload → Extract → Review → Approve → Post
Accounting platform	TallyPrime
Document inputs	Photos, scanned invoices, and PDFs

Objective: Reduce manual purchase-entry work while keeping a human review and approval step before any voucher is posted to TallyPrime.

Solution overview

The application accepts supplier bills as image files or PDFs, extracts invoice fields using OCR, validates the data, and presents a review screen. After an authorized user approves the draft, the integration sends a purchase voucher to TallyPrime and records the response and voucher reference.

Important: The bill image shared for discussion is blurry. Its invoice number, supplier details, and totals must be confirmed manually; this proposal does not assume unreadable values.

1. Proposed Architecture

Recommended first deployment: a local Windows application or trusted-network web dashboard, with the TallyPrime integration kept on the accounting computer or authorized office network.

Layer	Suggested technology	Responsibilities
User interface	React or simple web dashboard	Upload bills, edit fields, approve/reject, track status
Backend API	Python + FastAPI	Workflow, permissions, validation, and processing
OCR	PaddleOCR or Tesseract; optional cloud OCR	Read printed text from images and scanned PDFs
Extraction	Python parsing; optional AI model	Convert OCR output into structured invoice fields
Database	SQLite for prototype; PostgreSQL for production	Drafts, mappings, statuses, audit events
Tally connector	TallyPrime XML over configured HTTP interface	Send approved voucher and process Tally response
Document storage	Access-controlled local folders	Retain original bills and reviewed records

Workflow

- 1. Upload a photo, scanned invoice, or PDF.
- 2. Improve image quality where possible and run OCR.
- 3. Extract supplier, invoice number/date, item rows, quantities, rates, taxes, and total.
- 4. Map the supplier to an existing Tally ledger and map items to stock items or purchase ledgers.
- 5. Validate required fields, totals, applicable GST, and possible duplicates.
- 6. Display the source bill and editable extracted data to the reviewer.
- 7. On approval, send the voucher to TallyPrime and save the response and voucher reference.

2. Required Modules

A. Bill Upload & Document Management

- JPG, PNG, PDF, and scanned invoice upload.
- Unique processing ID, original-file retention, and batch upload.

B. OCR & Invoice Extraction

- Supplier, invoice number, invoice date, GSTIN, item descriptions, quantities, rates, discounts, tax amounts, and total.
- Confidence indicators and manual correction for blurry or low-confidence fields.

C. Review & Validation Dashboard

- Side-by-side original bill and editable extracted fields.
- Supplier/item mapping, total reconciliation, tax checks, duplicate detection, approve/reject/send-back.

D. TallyPrime Integration

- Use the configured TallyPrime interface to create purchase vouchers after approval.
- Handle successful imports, missing masters, invalid fields, and failed posting responses.

E. Tracking & Audit

- Statuses: Pending, Needs Correction, Approved, Posted, and Failed.

- Record reviewer, timestamps, changes, source document, Tally response, and voucher reference.

3. Review Screen & Voucher Rules

The review screen should show the source bill beside the extracted data. The operator must be able to correct fields before approval; the system must not post automatically in the initial workflow.

Field group	Fields to review	Validation
Supplier/invoice	Supplier ledger, invoice number, invoice date, GSTIN	Required fields; ledger mapping; duplicate check
Line items	Description, stock item/ledger, quantity, unit, rate, discount	Item mapping and line amount calculation
Tax/total	Taxable value, CGST/SGST/IGST where applicable, total	Reconcile invoice total within configured rounding tolerance
Posting	Voucher date, voucher type, narration, company	Company/configuration check and successful Tally response

Supported purchase scenarios

- **Accounting purchase:** allocate to the appropriate purchase ledger and applicable tax ledgers, with supplier liability recorded according to the company's configuration.
- **Inventory purchase:** include stock items, units, quantities, rates, purchase ledger allocation, applicable taxes, and supplier liability.

The correct voucher format depends on the company's accounting configuration and whether it uses accounting or inventory invoice mode. Confirm the required XML fields against a sample purchase voucher exported from the actual TallyPrime company.

4. Validation & Safety Controls

- Require supplier ledger mapping and invoice number/date before approval.
- Check for duplicates using company, supplier, invoice number, and date; also check existing Tally records where feasible.
- Reconcile line values, discounts, tax amounts, and invoice total with a configured rounding tolerance.
- Do not treat OCR output as authoritative for tax classification or accounting treatment; require review when uncertain.
- Keep the Tally endpoint on the local machine or trusted network; do not expose it directly to the public internet.
- Restrict approval/posting actions to authorized users and maintain an audit trail.
- Use idempotent posting/retry controls so a network timeout does not cause a second voucher.
- Mark a bill Posted only after a successful response and reconciliation of the returned voucher reference.
- Back up the test company before integration testing.

5. Prototype Plan

Schedule	Work	Deliverable
Days 1–2	Project setup, upload, storage, database	Application skeleton
Days 3–5	OCR and structured field extraction	Extracted invoice draft
Days 6–8	Review dashboard, mappings, validation, approval	Review workflow
Days 9–12	TallyPrime connector, test voucher, error handling	Test-company posting
Days 13–15	Invoice-format tests, duplicate/retry tests, handover	Test report and handover

Estimated duration: 10–15 working days for a first prototype, assuming timely access to a test TallyPrime company, sample invoices, and required configuration. This is a planning estimate, not a fixed delivery commitment.

6. Prototype Acceptance Checklist

- Upload a bill image and PDF.
- Extract supplier, date, invoice number, line items, and amount.
- Edit incorrect OCR fields before approval.
- Map supplier ledger and purchase/stock items.
- Validate applicable tax and invoice totals.
- Prevent duplicate posting on retry.
- Post an approved voucher to a test TallyPrime company.
- Show a posting reference or a readable error.
- Retain the original invoice and approval audit log.

7. Inputs Needed Before Live Integration

- Exact TallyPrime release and whether it runs locally, on an office network, or in a hosted environment.
- A test company or authorized test environment; do not begin with live accounts.
- Sample purchase vouchers exported from TallyPrime and representative supplier bills, including clear scans.
- Confirmation of accounting-ledger-only versus inventory purchase entry, and the company's GST configuration.
- Approval roles and rules for low-confidence OCR, missing ledgers, and tax discrepancies.

Recommended first release
Local Windows deployment • Human review before posting • Test-company integration • OCR confidence warnings • Duplicate protection • Audit trail

Prepared as a solution design. No live Tally connection or voucher posting has been performed.